

Leakage-controlled and deployment-aware input-resolution selection for dense broiler detection in precision poultry monitoring

Authors: Yanqi Song¹, Jianxiong Xu^{1*}, Huipin Lin², Zhekang Dong^{2*}

Affiliations: ¹ Hunan University of Technology, Zhuzhou, Hunan, China; ² Hangzhou Dianzi University, Hangzhou, Zhejiang, China

Corresponding authors: Jianxiong Xu, xujianxiong8411@163.com; Zhekang Dong, englishp@hdu.edu.cn

Abstract

Purpose Input resolution is a critical design variable for vision-based broiler monitoring, but its apparent benefit can be distorted by data leakage, source imbalance, counting-threshold calibration, and deployment cost. This study developed a leakage-controlled and deployment-aware protocol for selecting input resolution in dense poultry detection.

Methods A processed PIO broiler dataset with 1,487 images and 327,288 annotated instances was used as a high-density poultry-monitoring case study. A grouped train/validation/test split was constructed using exact-duplicate removal, source-group binding, density-aware allocation, and cross-split leakage audits. YOLO11n was trained and evaluated at 800, 960, and 1280 pixels under the same split, architecture, and training configuration. Candidate resolutions were compared using global detection accuracy, scale- and density-stratified AP, validation-calibrated counting error, paired-bootstrap uncertainty, training time, GFLOPs, latency, and memory pressure.

Results Test mAP₅₀₋₉₅ increased from 0.7199 at 800 pixels to 0.7420 at 960 pixels and 0.7543 at 1280 pixels, making 1280 pixels the accuracy-oriented setting. However, validation-calibrated counting selected 800 pixels as the lowest-MAE setting, with a test MAE of 12.30 instances per image. Under the stated multi-objective accuracy-cost weighting, 960 pixels was selected as the balanced candidate.

Conclusion Input resolution should not be reported as a universal best setting in dense broiler detection. It should be selected according to the intended monitoring objective: box-level detection accuracy, scalar counting accuracy, deployment-constrained monitoring, or an explicitly weighted combination of these objectives.

Keywords: Precision poultry monitoring; Broiler chicken detection; Input resolution; Leakage-controlled evaluation; Deployment-aware detection; Counting calibration

Introduction

Vision-based sensing is increasingly important in precision livestock and poultry monitoring because it provides non-contact information on animal distribution, flock density, resource use, and production-environment status (Guo et al. 2023; Guo et al. 2025; Li 2025; Boniche et al. 2026; Bhujel et al. 2025). In broiler production, reliable image-based detection is a prerequisite for automated counting, density assessment, welfare-related observation, and real-time monitoring. Commercial broiler scenes are nevertheless difficult: birds are visually similar, densely distributed, often occluded, and frequently represented by small bounding boxes under overhead or oblique views. These properties make detection

and counting sensitive to input sampling resolution, feature-map representation, non-maximum suppression, and computational constraints.

Input resolution is a direct but often under-diagnosed factor in dense small-object detection. Higher resolution can preserve more spatial detail and improve localization for small or crowded objects, but it also increases computational cost, GPU memory use, and training time. Moreover, resolution gains are not necessarily uniform across object scales, scene densities, or downstream objectives. A resolution that improves global mean average precision may not minimize scalar counting error or meet deployment constraints. Resolution selection should therefore be treated as a diagnostic and decision problem rather than as a simple hyperparameter search.

Evaluation leakage is a second critical issue. Dense image datasets collected from similar camera viewpoints, video streams, or repeated acquisition sessions may contain duplicate, near-duplicate, or source-related samples across training, validation, and test splits. If such leakage is not controlled, test performance may be overly optimistic and may not reflect generalization to new acquisition conditions. Similarity-aware splitting, perceptual hashing, SSIM-based near-duplicate auditing, and paired-bootstrap uncertainty estimation provide practical tools for making resolution comparisons more reliable (Wang et al. 2004; Biswas and Blanco-Medina 2021; Efron and Tibshirani 1993; Joeres et al. 2025).

This study proposes a leakage-controlled and deployment-aware input-resolution selection protocol for dense broiler detection. The protocol treats input resolution as a decision variable whose interpretation depends on the evaluation split, target metric, object scale, scene density, counting threshold, uncertainty, and computational constraint. Rather than asking whether one resolution is universally best, it asks how a resolution should be selected after the task objective and leakage-control requirements are explicitly defined.

The protocol combines MD5-based duplicate removal, source-group binding, density-aware grouped allocation, source-component diagnostics, and cross-split leakage audits before resolution effects are measured. YOLO11n is then trained at 800, 960, and 1280 pixels using a fixed split and fixed training configuration. These three resolutions are controlled candidate operating points, not an exhaustive search grid and not evidence for an inherent knee point.

The main contribution is a reproducible resolution-selection workflow rather than a new detector architecture or a universal recommendation for one input size. First, an auditable PIO split is constructed using exact-duplicate removal, source-group binding, density-aware grouped allocation, source-component reporting, and cross-split leakage audits. Second, candidate resolutions are evaluated with complementary detection, scale, density, counting, uncertainty, and deployment-cost diagnostics. Third, an explicit task-objective scoring and weight-sensitivity procedure is used to select a resolution according to the intended monitoring objective. In the tested PIO case, 1280 pixels is the accuracy-oriented setting, 800 pixels provides the lowest-cost baseline and the most favorable validation-calibrated counting setting, and 960 pixels is interpreted only as a balanced candidate under the stated multi-objective weighting.

Accordingly, the manuscript should be read as a resolution-selection workflow: construct an auditable split, evaluate candidate resolutions under fixed conditions, decompose their behavior by accuracy, counting, uncertainty, and cost, and then apply a stated task objective or weight-sensitivity analysis. This framing deliberately avoids the claim that 960 pixels is an inherent optimum.

The remainder of this paper is organized as follows. The next section reviews related work on poultry and livestock vision, dense small-object detection, leakage-controlled evaluation, and deployment-aware YOLO-style detection. The following section describes the PIO broiler-monitoring dataset, leakage-controlled split protocol, detector configuration, resolution-selection framework, and evaluation metrics. The results section reports global, scale-stratified, density-stratified, counting, uncertainty, deployment, cross-detector, and external directional-demonstration analyses. The discussion then interprets the implications of resolution-dependent behavior for precision poultry monitoring systems, followed by the conclusions.

Related work

Poultry and livestock vision in dense monitoring scenes

Computer vision has become an important sensing modality in precision livestock and poultry monitoring because it can support non-contact observation of animal distribution, activity, resource use, and flock-level production status (Guo et al. 2023; Guo et al. 2025; Li 2025; Boniche et al. 2026; Bhujel et al. 2025). In commercial broiler houses, image-based detection is a prerequisite for automated counting, density assessment, welfare-related observation, and real-time decision support. However, poultry scenes are more challenging than many conventional agricultural vision tasks because animals are visually similar, densely distributed, partially occluded, and often captured under variable illumination and camera viewpoints. These conditions make detection performance sensitive to object scale, scene density, annotation quality, and deployment constraints.

Previous broiler and livestock studies have used YOLO-style detection, attention mechanisms, density-map counting, and tracking tools to address animal detection, behavior monitoring, flock counting, and welfare-related analysis (Guo et al. 2023; Guo et al. 2025; Neethirajan 2022; van der Eijk et al. 2022; Lv et al. 2023; Guo et al. 2022). Public datasets and benchmark resources for precision poultry and livestock farming have also improved the reproducibility of vision-based monitoring studies (Li 2025; Boniche et al. 2026; Bhujel et al. 2025). The PIO dataset provides a large-scale open benchmark for broiler chicken detection under real poultry farming conditions (Boniche et al. 2026). Compared with prior studies that mainly focus on model design, behavior recognition, or counting architecture, the present study focuses on a different but practically important question: how input resolution should be selected for dense broiler detection when data leakage, scene density, counting accuracy, uncertainty, and deployment cost are jointly considered.

Dense small-object detection and input-resolution effects

Dense broiler detection can also be viewed as a dense small-object detection problem. General object detection has progressed from region-proposal methods to efficient one-stage and multi-scale detectors (Ren et al. 2017; Liu et al. 2016; Lin et al. 2014; Lin et al. 2017; Tan et al. 2020; Bodla et al. 2017; Padilla et al. 2021). However, dense small-object detection remains difficult because small targets provide limited spatial evidence, are easily confused with background clutter, and are strongly affected by occlusion, scale variation, and non-maximum suppression behavior (Nikouei et al. 2025; Feng et al. 2023; Cheng et al. 2022). In high-density poultry images, these challenges are amplified because hundreds of visually similar animals may appear in a single image, and many individual birds occupy only a limited number of pixels.

Multi-scale feature pyramids, compound scaling, slicing-aided inference, and high-resolution imagery benchmarks have been widely used to improve representation and localization for small objects (Lin et al. 2017; Tan et al. 2020; Akyon et al. 2022; Zhu et al. 2018; Li et al. 2020). These studies motivate the need to analyze not only global AP, but also scale-stratified and density-stratified behavior when changing input resolution. The present study differs from slicing-aided or multi-scale inference methods that modify how images are processed at inference time. Instead, the detector architecture, training configuration, data split, and test rules are kept fixed, and input resolution is treated as a controlled design variable. This allows resolution-dependent gains to be interpreted in terms of detection accuracy, counting behavior, uncertainty, and deployment cost rather than being reported as a simple hyperparameter search result.

Leakage-controlled evaluation, similarity audit, and uncertainty

Reliable evaluation is essential for precision agriculture vision systems because over-optimistic performance estimates can lead to incorrect deployment decisions. In image datasets collected from similar camera viewpoints, video streams, repeated sampling conditions, or source-related acquisition sessions, duplicate, near-duplicate, or source-related samples may appear across training, validation, and test subsets. If such leakage is not controlled, the test set may no longer represent truly unseen data, and reported detection or counting performance may be inflated (Joeres et al. 2025).

Image-based leakage control can be supported by exact-duplicate removal, source-group binding, perceptual-hash candidate screening, and SSIM-based near-duplicate verification (Wang et al. 2004; Biswas and Blanco-Medina 2021). For dense poultry scenes, source-level grouping is especially important because images collected from the same house, camera, or acquisition sequence may share strong visual similarity even when they are not exact duplicates. In addition, paired bootstrap analysis provides a practical way to quantify uncertainty in resolution-dependent AP differences while preserving image-level coupling among dense predictions (Efron and Tibshirani 1993). Therefore, a resolution-selection study should combine leakage-controlled split construction, cross-split similarity audit, and uncertainty analysis before drawing conclusions about the benefit of a particular input size.

YOLO-style detection and deployment constraints for precision poultry monitoring

YOLO-style detectors are widely used in real-time object detection because they provide a practical balance between accuracy and speed (Redmon et al. 2016; Redmon and Farhadi 2017, 2018; Bochkovskiy et al. 2020; Wang et al. 2021, 2023; Wang C.-Y. et al. 2024; Wang A. et al. 2024; Jocher and Qiu 2024). This balance is particularly relevant for precision poultry monitoring, where models may need to operate under limited GPU memory, edge-computing resources, or near-real-time farm-monitoring requirements. Later YOLO variants improved feature aggregation, training strategies, and detector efficiency, while recent software implementations such as Ultralytics YOLO11 further support practical deployment workflows (Jocher and Qiu 2024).

For dense broiler scenes, increasing input resolution can preserve more spatial detail and improve localization for small or crowded birds. However, higher resolution also increases GFLOPs, latency, memory pressure, and training time. Therefore, resolution selection should not be treated as a purely accuracy-oriented decision. A resolution that maximizes mAP may not be the best choice for scalar counting, low-cost deployment, or real-time monitoring. This motivates the leakage-controlled and deployment-aware resolution-selection protocol developed in this study.

Materials and methods

Overview of the leakage-controlled resolution-selection protocol

This study evaluated input resolution as a practical design variable for dense broiler detection in precision poultry monitoring. The goal was not to introduce a new detector architecture, but to define a reproducible workflow for choosing an input resolution under leakage-controlled evaluation and deployment-aware task objectives. The protocol separates the effect of input resolution from confounding factors such as duplicate or source-related data leakage, object-scale imbalance, scene-density variation, confidence-threshold calibration, uncertainty, and computational cost.

For an image I_i and an input resolution r in $\{800, 960, 1280\}$, the detector produces a set of bounding-box predictions $P_{i,r}$ with confidence scores and class labels. The detector architecture, training protocol, data split, and evaluation rules were held constant across all resolutions. Differences among candidate resolutions were therefore interpreted primarily as resolution-dependent effects rather than as consequences of changes in model capacity, optimization strategy, or dataset composition.

Fig. 1 summarizes the controlled front end of the proposed resolution-selection workflow. The workflow starts from the dense PIO broiler-monitoring dataset, applies duplicate removal, group-aware splitting, and density balancing to construct the leakage-controlled split, and then evaluates 800-, 960-, and 1280-pixel inputs with the same YOLO11n detector and experimental rules. The resulting outputs are interpreted in terms of detection performance, scale-density behavior, counting reliability, and deployment efficiency.

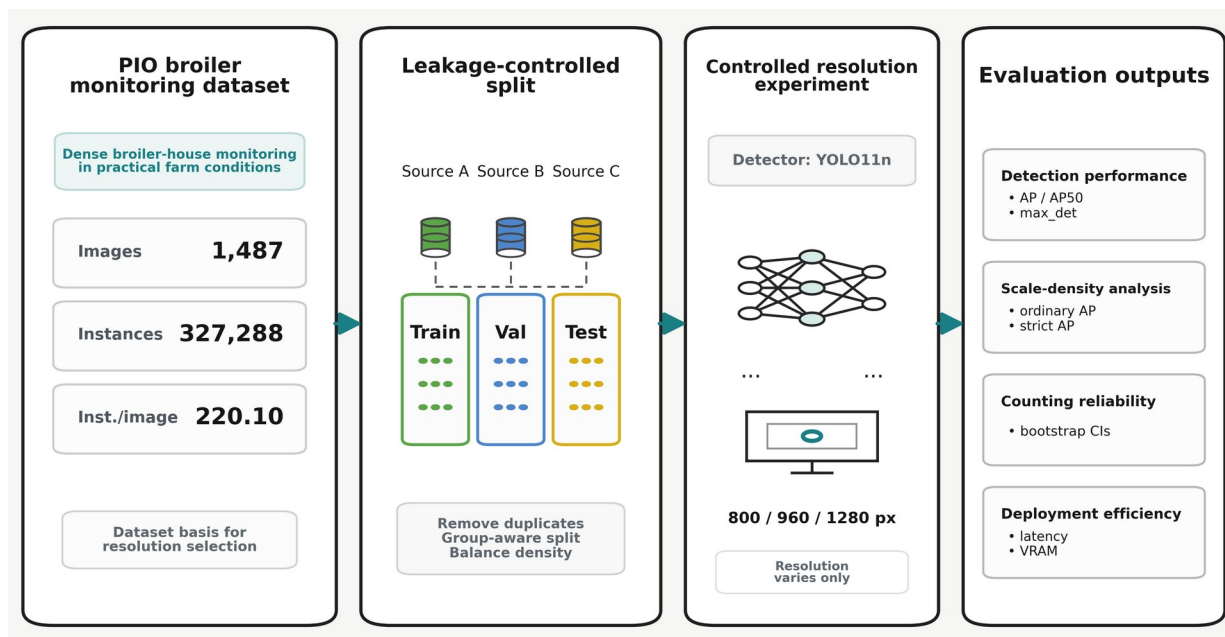


Fig. 1 Leakage-controlled workflow for input-resolution selection in dense broiler monitoring. The figure summarizes the PIO dataset basis, group-aware leakage-controlled split, controlled YOLO11n resolution experiment at 800, 960, and 1280 px, and the evaluation outputs used to compare detection performance, scale-density behavior, counting reliability, and deployment efficiency

Dense broiler-monitoring case study and leakage-controlled split

The PIO dataset (Boniche et al. 2026) was used as a real-world dense broiler-monitoring case study for precision poultry farming. The processed dataset contains 1,487 images and 327,288 annotated broiler instances, with an average of 220.10 instances per image. The fixed split, denoted as PIO-GRDB-MD5-7_1_2, contains 1,027 training images, 153 validation images, and 307 test images. The corresponding instance counts are 225,771, 36,905, and 64,612 for training, validation, and testing, respectively.

The original PIO folder names contain prefixes such as `train__` and `val__` inherited from the source collection; these prefixes do not reflect the leakage-controlled split assignment used in this study.

The split protocol was designed to reduce the risk of over-optimistic evaluation caused by duplicate images, near-duplicate images, or source-level overlap. Image-level MD5 hashing was first used to remove exact duplicates. Source groups were inferred from source-path and filename-pattern metadata and were treated as indivisible units during split construction so that near-contiguous or source-related samples were not distributed across training, validation, and test subsets. A density-balanced greedy allocation strategy was used to reduce severe distributional imbalance in the number of instances per image across subsets.

Four cross-split leakage checks were performed using original image paths, original file names, image-content MD5 values, and inferred source-group identifiers. In addition, near-duplicate screening was conducted using perceptual-hash candidates followed by structural similarity checks on 256 x 256 grayscale images (Wang et al. 2004). No confirmed cross-split duplicate or near-duplicate image pairs were found under the adopted audit criteria. The detailed leakage-control and near-duplicate audit configuration is summarized in Table 3, and the grouped split procedure is described in Algorithm 1.

Source-component statistics are reported in the main text (Table 2) to make the leakage-control claim directly auditable. At the source-component level, the final split contained 361 training components, 2 validation components, and 20 test components. The largest components contained 352, 130, and 151 images in the training, validation, and test subsets, respectively. These diagnostics expose source-group concentration and split imbalance transparently rather than treating the split as a random image-level partition. The validation subset was used for model monitoring and confidence-threshold calibration, whereas the test subset was reserved for performance reporting and diagnostic analysis.

Table 1 Dataset split statistics under the PIO-GRDB-MD5-7_1_2 protocol

Split	Images	Image ratio (%)	Instances	Instance ratio (%)	Mean inst./image
Train	1,027	69.07	225,771	68.98	219.84
Validation	153	10.29	36,905	11.28	241.21
Test	307	20.65	64,612	19.74	210.46
Total	1,487	100.00	327,288	100.00	220.10

Table 2 Source-component-level split diagnostics for the leakage-controlled split

Split	Source components	Images	Instances	Largest component images	Largest component instances
Train	361	1,027	225,771	352	150,312

Split	Source components	Images	Instances	Largest component images	Largest component instances
Validation	2	153	36,905	130	35,690
Test	20	307	64,612	151	29,961

Note: Source components were inferred from source-path and filename-pattern metadata and were kept within a single split. The table is reported in the main manuscript to make source-group binding, component concentration, and split imbalance transparent.

Table 3 Leakage-control and near-duplicate audit configuration

Step	Method	Criterion / threshold	Output
Exact-duplicate removal	Image-content MD5 hashing	Images with identical MD5 hashes were treated as exact duplicates	Duplicate-free processed image set
Source-group inference	Source-path and filename-pattern parsing	Images sharing the same inferred acquisition/source pattern were assigned to the same source group	Source-group identifier for each image
Source-group binding	Grouped split construction	All images within the same source group were kept in a single split	Reduced source-related cross-split leakage risk
Density-aware allocation	Greedy grouped allocation with density balancing	Groups were assigned while considering target split ratio and image-level instance counts	Train/validation/test split with controlled density distribution
Cross-split leakage audit	Leakage-key comparison	Source path, filename, image-content MD5, and inferred source group were checked across splits	PASS/FAIL leakage audit result
Near-duplicate candidate search	pHash/dHash screening	Candidate pairs were selected using pHash ≤ 6 or dHash ≤ 6	Candidate near-duplicate pairs
Near-duplicate verification	SSIM verification	SSIM ≥ 0.95 on 256 x 256 grayscale images	Confirmed near-duplicate pairs
Fixed split release	Fixed split manifest	PIO-GRDB-MD5-7_1_2 with train/validation/test subsets	Leakage-controlled split for all experiments

Note: Source groups were inferred from source-path and filename patterns and treated as indivisible units during split construction. The near-duplicate audit used perceptual-hash candidate search followed by SSIM verification.

Algorithm 1 Leakage-controlled grouped split construction

Input: processed image set I , annotation set A , target split ratio 7:1:2, source-path and filename metadata M , image-level density values, and leakage keys {source path, filename, image-content MD5, source group}. Output: leakage-controlled train, validation, and test split manifests.

1. Compute the image-content MD5 hash for each image and remove exact duplicates.
2. Infer a source-group identifier for each remaining image using source-path and filename-pattern metadata.
3. Construct grouped units so that all images in the same source group remain together.
4. For each group, compute the number of images, number of annotated instances, mean density, and density-bin composition.
5. Sort grouped units by size and density characteristics.
6. Assign grouped units to train, validation, or test using greedy allocation while minimizing deviations from the target image ratio, target instance ratio, and density-bin balance.
7. Compute split-level image, instance, and density statistics.

8. Perform cross-split leakage checks using source path, filename, image-content MD5, and source group.
9. Perform near-duplicate audit using pHash ≤ 6 or dHash ≤ 6 candidate screening followed by SSIM ≥ 0.95 verification on 256 x 256 grayscale images.
10. If any cross-split leakage or confirmed near-duplicate pair is found, revise the grouped allocation and repeat the audit.
11. Freeze the split manifest as PIO-GRDB-MD5-7_1_2.

Detector configuration for dense broiler detection

YOLO11n was selected as the fixed main detector for the controlled resolution-selection experiments (Jocher and Qiu 2024). A lightweight detector is appropriate for deployment-oriented dense small-object detection because inference speed, memory footprint, and training cost are practical constraints. The architecture was not modified; input resolution was the controlled variable. YOLOv8n was evaluated only as a cross-detector robustness check, not as a detector-ranking benchmark.

Three input resolutions were evaluated: 800, 960, and 1280 pixels. The 800-pixel setting represents the lowest-cost candidate, the 960-pixel setting is an intermediate candidate to be assessed under explicit task objectives, and the 1280-pixel setting represents a higher-accuracy, higher-cost configuration. All three resolutions are compatible with the detector stride and could be trained with the same batch size on the available GPU.

All primary PIO YOLO11n models were trained for 300 epochs with batch size 2, stochastic gradient descent optimization, cosine learning-rate scheduling, and pretrained weights. The main YOLO11n experiments used seed 42. To examine whether the observed resolution trend was caused by a single random initialization or training trajectory, additional YOLO11n seed-repeatability experiments were conducted for all three resolutions using seeds 123 and 2024. The same training split, validation split, augmentation pipeline, optimizer configuration, and evaluation protocol were maintained across resolutions.

Detection evaluation and maximum-detection sensitivity analysis

Standard detection evaluation was performed on the fixed test split using precision, recall, mAP50, and mAP50-95. Because each image may contain hundreds of instances, the maximum number of retained detections per image is a critical evaluation parameter. A low maximum-detection setting can suppress recall in high-density scenes and underestimate detection performance.

The main test evaluation therefore used max_det=1000. Sensitivity analysis with lower and higher caps was conducted to verify that the main results were not limited by the detection cap and that max_det=1000 was sufficient for the density range of the test split.

Scale- and density-stratified evaluation of resolution effects

To identify where resolution-related performance gains originate, the test split was analyzed using both object-scale and image-density groupings.

For scale-stratified diagnosis, ground-truth boxes in the test split were divided into four groups according to the quartiles of the square root of bounding-box area. These groups are denoted as tiny, small, medium, and large. This quartile-based design ensures that the number of instances

in each scale group is approximately balanced, preventing a dominant scale group from overwhelming the diagnostic analysis.

For density-stratified diagnosis, test images were divided into four groups according to the number of annotated instances in the full image. These groups are denoted as low, medium, high, and ultra-high density. The density group is defined at the full-image level, even when qualitative visualization only displays a local crop for readability.

To avoid ambiguity between subgroup-local evaluation and global contribution analysis, this study distinguishes ordinary subgroup AP from strict global-first diagnostic AP.

Ordinary subgroup AP is computed after filtering the evaluation set to a predefined subgroup, such as a target scale group or a density group. Predictions and ground-truth objects outside the subgroup are excluded before forming the precision-recall curve. Therefore, ordinary subgroup AP answers the question of how well the detector performs when that subgroup is evaluated as a separate subset.

In contrast, strict global-first diagnostic AP preserves the full-test prediction order before subgroup attribution. Predictions are ranked and matched globally, and matched true positives are then assigned to their predefined scale or density groups. The ranking is not re-optimized within each subgroup.

Strict global-first diagnostic AP therefore answers a complementary question: under the detector's global operating order, how much does a subgroup contribute to the overall ranked detection result? It is contribution-oriented, not numerically interchangeable with ordinary subgroup AP, and is used only as an auxiliary diagnostic metric. Additional implementation details and supporting tables are provided in Online Resource 2.

Bootstrap uncertainty analysis

Paired image-level bootstrap analysis was used to assess the stability of resolution-dependent performance differences (Efron and Tibshirani 1993). The image was used as the resampling unit because predictions and ground-truth annotations within the same image are statistically coupled, especially in dense scenes. For each bootstrap replicate, the same resampled image set was applied to all resolution settings, allowing paired comparisons between resolutions.

The bootstrap analysis estimated confidence intervals for AP differences between 800 and 960, between 960 and 1280, and between 800 and 1280. This paired design reduces the influence of image-composition variability and provides a more stable estimate of whether observed resolution gains are consistent across the test set.

Counting calibration for broiler detection outputs

Detection accuracy and counting accuracy are related but not equivalent in dense small-object applications. Average precision is ranking based, whereas image-level counting error depends on a fixed confidence threshold and score calibration. Counting performance was therefore evaluated separately from AP.

A validation-calibrated counting protocol was used. For each resolution, confidence thresholds were swept on the validation split, and the threshold with the lowest mean absolute error was selected, with

RMSE and absolute bias used as auxiliary criteria when needed. The selected threshold was then fixed and applied once to the test split. The reported counting metrics were MAE, RMSE, MAPE, bias, and relative error.

The validation subset contained two source components, with the largest component contributing 130 of 153 validation images. This concentration is reported transparently because it may affect threshold calibration. Sensitivity to threshold choice was additionally examined using the diagnostic test-set confidence sweep in Online Resource 3.

The test-set confidence sweep was diagnostic only and was not used to select the reported counting threshold. This separation prevents threshold selection from being tuned directly on the test set and provides a more deployment-like estimate of counting behavior.

Deployment-oriented benchmarking for precision poultry monitoring

Deployment-oriented benchmarking was conducted under the same hardware and software environment for all resolutions. All training and benchmarking experiments were performed on a single NVIDIA GeForce RTX 5060 Laptop GPU with 8 GB of dedicated GPU memory. The repeated locked-weight deployment benchmark used the validation split, batch size 2, workers=0, confidence threshold=0.001, NMS IoU=0.70, max_det=1000, GPU FP16 inference, five warm-up repeats, and thirty formal repeats. Median preprocessing time, inference time, postprocessing time, total latency, latency IQR, GFLOPs, and validation/inference peak CUDA memory were reported.

Peak memory in the deployment benchmark refers to validation/inference CUDA memory, not training peak VRAM. Training time was reported separately because high-resolution model development can be a major practical cost. The benchmark was designed to compare relative resolution costs under a fixed local environment, not to claim absolute latency across all edge devices or inference frameworks.

Hardware-specific acceleration platforms such as TensorRT, ONNX Runtime, embedded GPUs, and optimized NMS implementations may change absolute latency values. Therefore, the deployment analysis is interpreted together with training time, GFLOPs, total latency, and memory pressure rather than using a single speed metric as the only deployment criterion.

Reproducibility materials

To support reproducibility and transparent evaluation, a public reproducibility release is provided for this study (Song 2026). The release contains the leakage-controlled split manifest, source-group metadata, leakage-audit summaries, group definitions, bootstrap outputs, counting-calibration outputs, supplementary source data, evaluation scripts, traceability logs, and fixed figure source data.

The release also includes scripts for split construction, protocol-figure generation, plotting from fixed CSV source data, qualitative-figure metadata, strict diagnostic AP plotting, validation confidence-MAE sweeps, YOLOv8n robustness checks, locked-weight deployment benchmarking, multi-objective scoring, weight-sensitivity analysis, VisDrone2019-DET conversion, and VisDrone split-audit reporting. Raw PIO and VisDrone images, trained model weights, and prediction overlays are not redistributed; they must be obtained or generated from the original sources and local prediction files.

Results

Global detection performance across input resolutions

The standard test-set results are summarized in Table 4. The evaluation used the fixed test split, confidence threshold of 0.001, NMS IoU of 0.70, and max_det=1000.

Table 4 Standard test-set detection performance with max_det = 1000

Resolution	Precision	Recall	mAP50	mAP50-95
800	0.9301	0.9283	0.9643	0.7199
960	0.9314	0.9385	0.9739	0.7420
1280	0.9350	0.9438	0.9741	0.7543

Detection performance improved as input resolution increased. Test mAP50-95 rose from 0.7199 at 800 pixels to 0.7420 at 960 pixels and 0.7543 at 1280 pixels. The point-estimate gain from 800 to 960 was +0.0221, whereas the gain from 960 to 1280 was +0.0123, suggesting diminishing returns. The mAP50 values at 960 and 1280 pixels were nearly saturated (0.9739 and 0.9741), indicating that the additional benefit of 1280 pixels was mainly associated with stricter localization quality rather than coarse IoU=0.50 detection.

Bootstrap analysis (Table S12b) supports this interpretation. The mean AP50 gain from 960 to 1280 pixels was +0.000865, and its 95% confidence interval included zero [-0.000187, 0.001886]. Thus, higher resolution primarily improved localization across stricter IoU thresholds rather than coarse detection at IoU=0.50. The main resolution-diagnostic conclusions are therefore drawn from mAP50-95, with AP50 treated as a saturation indicator.

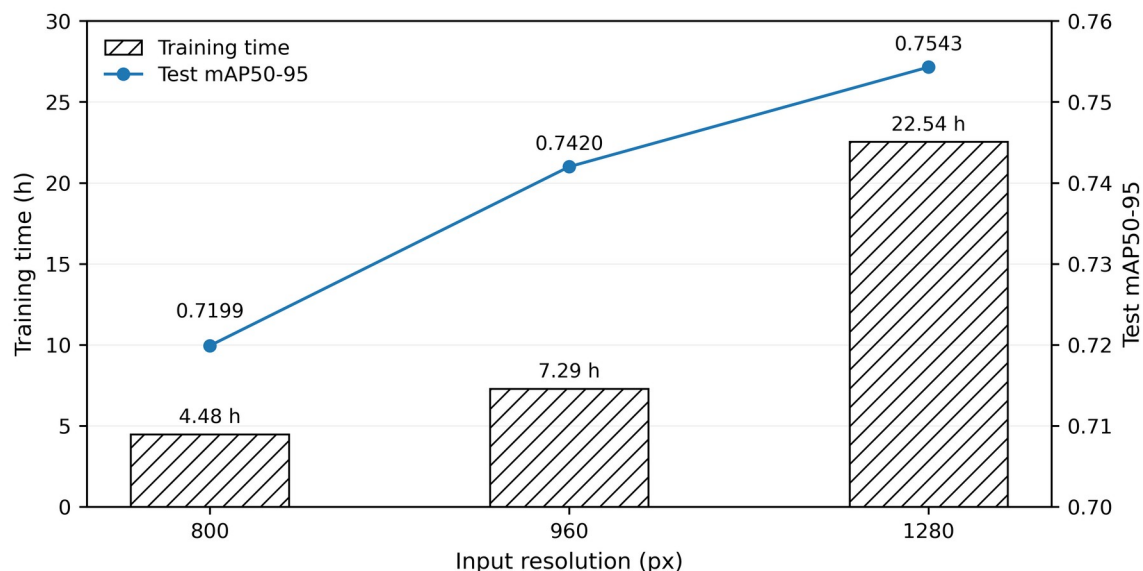


Fig. 2 Training time and test mAP50-95 across input resolutions. Bars show total training time, and the line shows the corresponding test mAP50-95.

Maximum-detection sensitivity in high-density poultry scenes

The sensitivity of mAP50-95 to the maximum number of retained detections is reported in Table 5. Because the test images may contain hundreds of instances, a low maximum-detection cap can underestimate dense-scene performance.

Table 5 Maximum-detection sensitivity summarized by mAP50-95

Resolution	mAP50-95 @300	mAP50-95 @1000	mAP50-95 @1500	Delta 1000-300	Delta 1500-1000
800	0.6733	0.7199	0.7199	+0.0466	0.0000
960	0.6925	0.7420	0.7420	+0.0495	0.0000
1280	0.7025	0.7543	0.7545	+0.0518	+0.0002

The setting max_det=300 substantially underestimated performance in high-density images. Increasing max_det from 300 to 1000 improved mAP50-95 by +0.0466, +0.0495, and +0.0518 for 800, 960, and 1280 pixels, respectively. Further increasing max_det from 1000 to 1500 produced negligible changes. Thus, max_det=1000 was sufficient for the main evaluation while avoiding an unnecessarily high detection cap.

Scale-stratified resolution effects

The scale-stratified results are reported in Table 6 and visualized in Fig. 3. Ordinary scale-stratified AP treats the target scale group as positives and ignores ground-truth boxes from other scale groups.

Table 6 Scale-stratified AP across input resolutions

Scale group	Images	Instances	800	960	1280
tiny	279	16,153	0.4774	0.5157	0.5239
small	300	16,153	0.5912	0.6263	0.6341
medium	289	16,153	0.6959	0.7162	0.7246
large	278	16,153	0.7585	0.7634	0.7863

Note: The Images column reports the number of test images that contain at least one object in the target scale group; because a single image can contain objects across multiple scale groups, the sum exceeds the total number of test images.

Detection performance increased with object scale. The tiny group had substantially lower AP than the large group, confirming that small objects remain the most challenging cases in dense broiler scenes. Increasing input resolution from 800 to 960 improved all scale groups. Increasing from 960 to 1280 produced smaller but still positive gains, with the large group showing the most notable improvement in that interval.

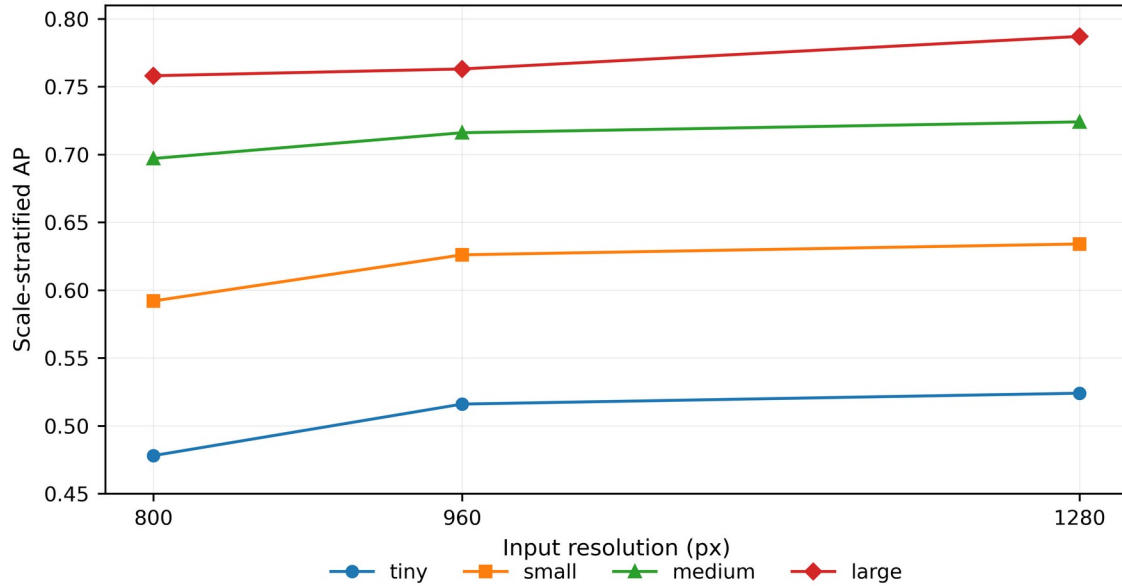


Fig. 3 Scale-stratified AP across input resolutions. Ground-truth boxes from the target scale group were treated as positives, and ground-truth boxes from other scale groups were ignored.

Density-stratified resolution effects

The density-stratified results are summarized in Table 7 and illustrated in Fig. 4. Density groups were defined by the number of ground-truth instances in the full test image.

Table 7 Density-stratified AP across input resolutions

Density group	Images	Instances	Mean inst./image	800	960	1280
low	78	7,096	90.97	0.6446	0.6544	0.6583
medium	76	12,300	161.84	0.7407	0.7402	0.7584
high	76	17,924	235.84	0.7341	0.7516	0.7679
ultra-high	77	27,292	354.44	0.7289	0.7570	0.7702

Under ordinary density-subset AP, 1280 pixels achieved the highest AP in all four density groups. In the high-density group, mAP50-95 increased from 0.7341 at 800 pixels to 0.7516 at 960 pixels and 0.7679 at 1280 pixels. In the ultra-high-density group, it increased from 0.7289 to 0.7570 and 0.7702. These results indicate that, under conventional subset evaluation, higher resolution was beneficial across density groups.

Strict global-first density diagnostic AP provides a complementary contribution-oriented view. In the high-density group, strict diagnostic mAP50-95 increased from 0.2299 at 800 pixels to 0.2622 at 960 pixels, whereas the 1280-pixel value was 0.2593. In the ultra-high-density group, the corresponding values were 0.2940, 0.3170, and 0.3170. These values should not be compared numerically with ordinary density-subset AP because the strict metric preserves the global ranked prediction list and does not re-optimize the ranking within each density group. The strict diagnostic result therefore does not imply that

1280 pixels performs worse in high-density scenes; standard global mAP and ordinary density AP still identify 1280 pixels as the highest-accuracy setting.

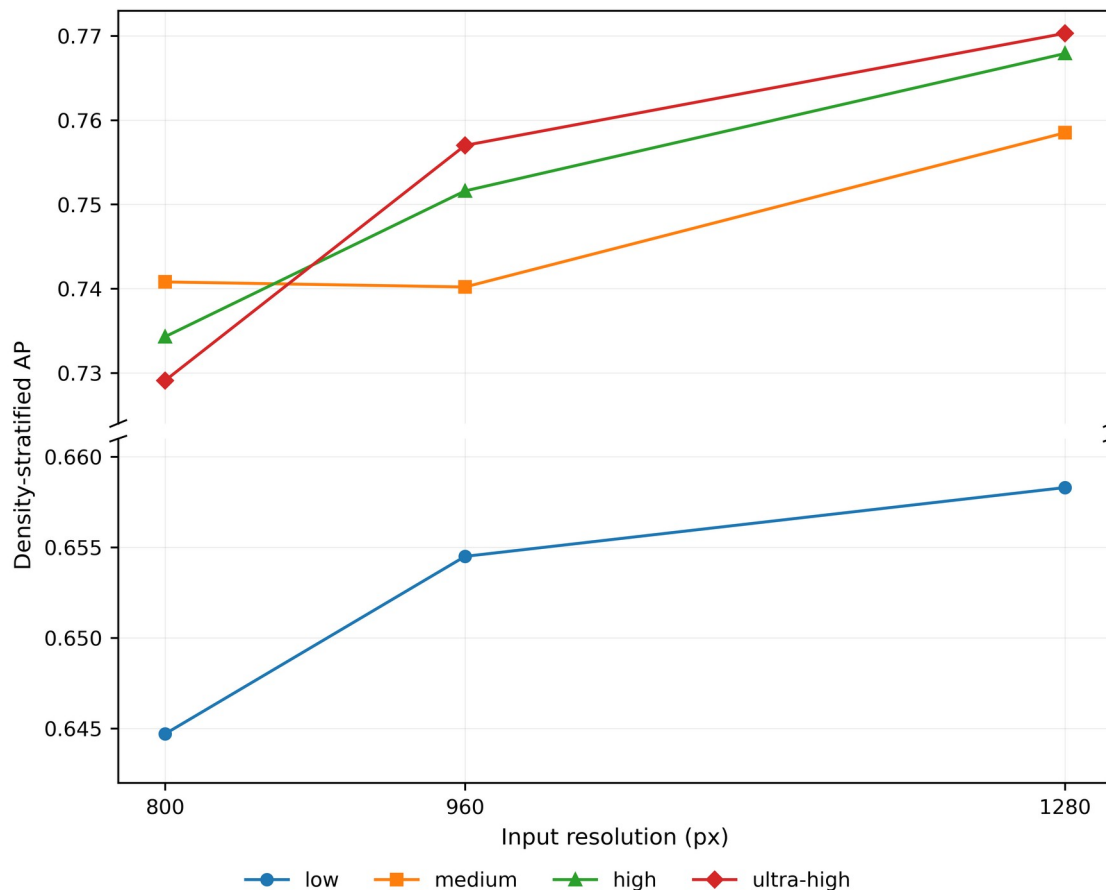


Fig. 4 Density-stratified AP across input resolutions. Density groups were defined by the number of ground-truth instances in the full test image. The y-axis is broken only to improve visualization of the low-density group and the three higher-density groups; the plotted AP values are unchanged.

TP/FN/FP diagnostics and representative dense-scene visualization

The TP/FN/FP diagnostics further clarify the density-stratified behavior. In high- and ultra-high-density groups, 960 pixels reduced false negatives relative to 800 pixels and improved recall. In the high-density group, FN decreased from 621 to 568. In the ultra-high-density group, FN decreased from 893 to 680. From 960 to 1280 pixels, the TP/FN gains were smaller, but false positives were further reduced. In the ultra-high-density group, TP/FN values were similar for 960 and 1280 pixels (26,612/680 and 26,605/687), while FP decreased from 3,579 to 3,079. The additional benefit of 1280 pixels in dense scenes was therefore more related to precision and false-positive reduction than to a large recall increase.

Representative qualitative results are shown in Fig. 5. The visualization compares predictions under different input resolutions in high-density and ultra-high-density scenes. Yellow boxes indicate ground-truth annotations, and cyan boxes indicate predictions. For readability, only predictions with confidence

≥ 0.50 are displayed; TP, FN, and FP values were computed using confidence threshold 0.25 and IoU threshold 0.50.

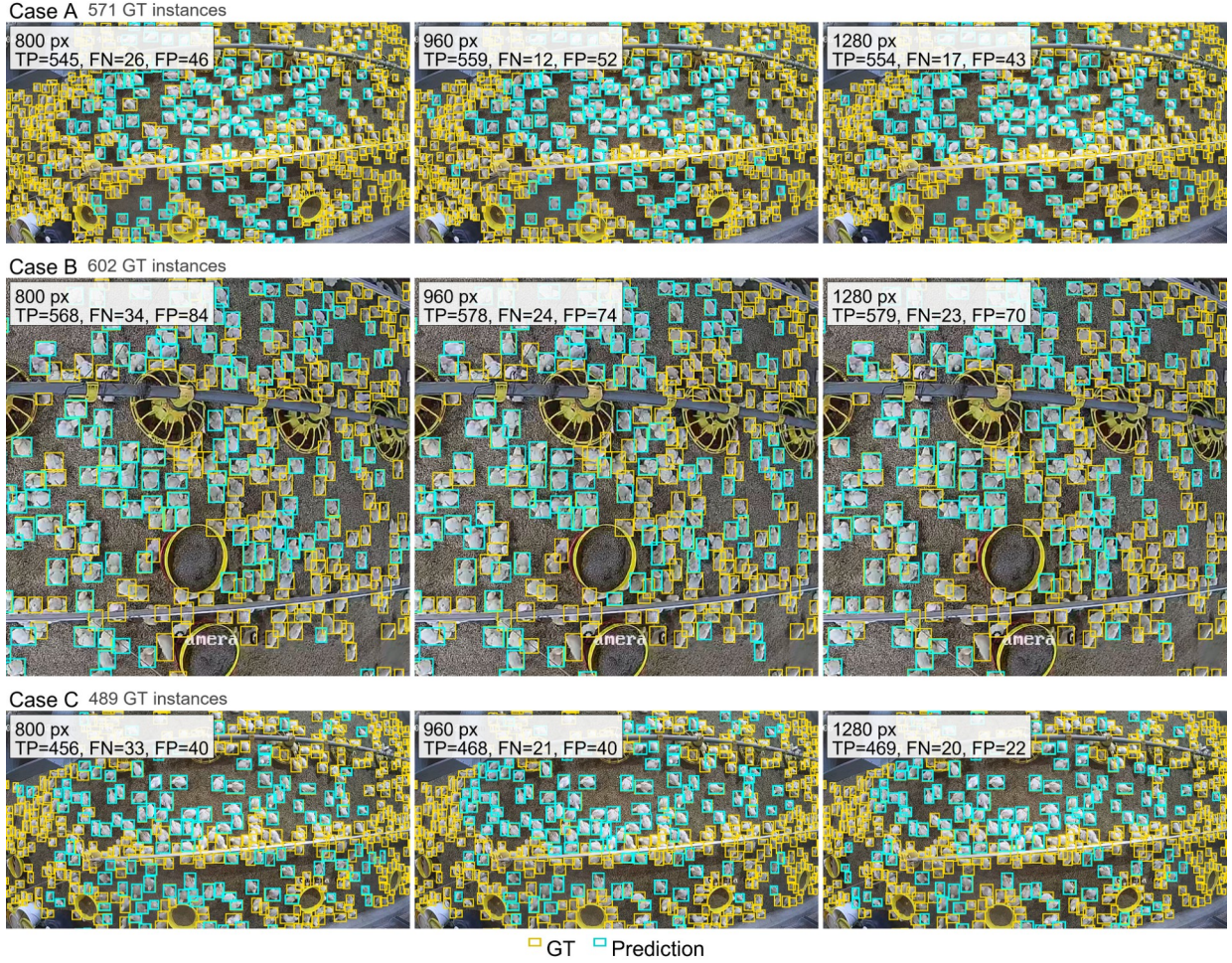


Fig. 5 Qualitative comparison of YOLO11n predictions under different input resolutions in representative dense-scene crops. Yellow boxes indicate ground-truth annotations, and cyan boxes indicate predictions. TP, FN, and FP were computed at confidence threshold = 0.25 and IoU threshold = 0.50, while only predictions with confidence ≥ 0.50 are displayed for readability. Local dense crops are shown to make the box-level differences visible. Full-image density labels, crop metadata, and visualization source data are provided in the reproducibility package. Filename prefixes such as train__ and val__ were inherited from the original PIO folders and do not indicate the split assignment used in this study

Bootstrap uncertainty of resolution gains

The paired bootstrap confidence intervals are reported in Table 8. The analysis used image-level paired resampling with 1000 bootstrap replicates.

Table 8 Paired image-level bootstrap confidence intervals for mAP50-95 differences

Comparison	Mean delta mAP50-95	95% CI
960 - 800	+0.0195	0.0167-0.0221
1280 - 960	+0.0136	0.0113-0.0157
1280 - 800	+0.0330	0.0305-0.0355

The confidence intervals did not cross zero, indicating stable mAP50-95 improvements for both resolution increases. The estimated gain from 1280 over 960 pixels was smaller than the gain from 960 over 800 pixels, supporting diminishing marginal returns. The bootstrap delta-difference analysis is provided in Online Resource 5.

Counting calibration from detection outputs

The test-set counting results obtained using validation-calibrated thresholds are reported in Table 9. Detection AP and counting error are not equivalent because AP is a ranking-based metric, whereas counting depends on a fixed confidence threshold.

Table 9 Test counting performance using validation-calibrated confidence thresholds

Resolution	Val-selected conf	GT total	Pred total	MAE	RMSE	MAPE (%)	Bias	Rel. err. (%)
800	0.45	64,612	64,735	12.30	18.31	7.33	+0.40	+0.19
960	0.50	64,612	62,333	15.72	22.26	8.79	-7.42	-3.53
1280	0.50	64,612	63,215	14.01	20.49	7.98	-4.55	-2.16

The validation-selected confidence thresholds were 0.45, 0.50, and 0.50 for 800, 960, and 1280 pixels, respectively. When these thresholds were applied to the test split, the 800-pixel model obtained the lowest MAE (12.30), followed by 1280 pixels (14.01) and 960 pixels (15.72). This result is central to the manuscript: the preferred input resolution changes with the task objective. The 800-pixel counting result does not contradict the AP results; it reflects the interaction between score calibration, validation-selected thresholds, and the conversion of ranked detections into scalar counts. At the validation-selected threshold of 0.50, the 960- and 1280-pixel models produced conservative total predicted counts, whereas the 800-pixel threshold of 0.45 produced near-zero total-count bias. Consequently, a counting-only deployment objective may select 800 pixels, a box-level detection objective may select 1280 pixels, and a joint accuracy-cost objective may select a different candidate through the explicit multi-objective rule.

The diagnostic test-set confidence sweep in the supplementary material further shows that test-tuned thresholds reduce the MAE of 960- and 1280-pixel models, but 800 pixels remains highly competitive for pure counting. The counting results are therefore a deliberate stress test of any single-resolution conclusion and support the claim that resolution selection must be tied to scalar counting, ranked box-level detection, or a weighted accuracy-cost objective.

Latency, memory, and training-cost trade-off

The deployment-oriented benchmark and task-objective-dependent resolution-selection summary are shown in Table 10. The locked-weight benchmark used the validation split, batch size 2, workers=0, confidence threshold=0.001, NMS IoU=0.70, max_det=1000, GPU FP16 inference, five warm-up repeats, and thirty formal repeats. The multi-objective score combined main test mAP50-95 with training-time efficiency, GFLOPs efficiency, latency efficiency, and memory efficiency. All normalization terms are defined below for the candidate resolution set $R = \{800, 960, 1280\}$.

$$N_{\text{acc}}(r) = [A(r) - \min(A)] / [\max(A) - \min(A)]. \quad (1)$$

$$N_{\text{cost},c}(r) = [\max(C_c) - C_c(r)] / [\max(C_c) - \min(C_c)], \quad c \in \{\text{training time, GFLOPs, latency, memory}\}. \quad (2)$$

$$\text{Score}(r) = 0.50 N_{\text{acc}}(r) + 0.20 N_{\text{train}}(r) + 0.10 N_{\text{GFLOPs}}(r) + 0.10 N_{\text{latency}}(r) + 0.10 N_{\text{memory}}(r). \quad (3)$$

Additional counting-calibration sensitivity analyses

Two additional sensitivity analyses were added to clarify whether the validation-calibrated counting conclusion depends on validation-source composition or confidence-threshold protocol. First, the validation split was partitioned by source component and each component was used independently to select the confidence threshold. Second, fixed-threshold, validation-calibrated, and diagnostic test-set best protocols were compared on the same held-out test split. These analyses are used to interpret the stability of the counting conclusion and are not alternative model-selection rules.

Table 9a Source-component sensitivity of validation-calibrated counting thresholds

Resolution	Calibration subset	Images	Selected conf.	Test MAE	Test RMSE	Bias	Rel. total err. (%)
800	Full validation	153	0.45	12.30	18.31	+0.40	+0.19
800	Component 1 only	130	0.45	12.30	18.31	+0.40	+0.19
800	Component 10 only	23	0.40	12.44	19.57	+6.34	+3.01
960	Full validation	153	0.50	15.72	22.26	-7.42	-3.53
960	Component 1 only	130	0.50	15.72	22.26	-7.42	-3.53
960	Component 10 only	23	0.35	15.22	22.36	+10.70	+5.09
1280	Full validation	153	0.50	14.01	20.49	-4.55	-2.16
1280	Component 1 only	130	0.50	14.01	20.49	-4.55	-2.16
1280	Component 10 only	23	0.45	12.89	19.23	+0.63	+0.30

The full-validation and dominant-component calibrations selected the same thresholds for all three resolutions. Calibration on the smaller component changed the selected thresholds, indicating that validation-source composition can influence scalar counting calibration. Under the formal full-validation protocol, 800 pixels retained the lowest test MAE.

Table 9b Counting-threshold robustness under fixed, validation-calibrated, and diagnostic test-set thresholds

Resolution	Protocol	Selected conf.	MAE	RMSE	MAPE (%)	Bias	Rel. total err. (%)
800	Fixed conf. 0.25	0.25	26.07	34.43	13.72	+25.66	+12.19
800	Validation calibrated	0.45	12.30	18.31	7.33	+0.40	+0.19
800	Test diagnostic best	0.45	12.30	18.31	7.33	+0.40	+0.19
960	Fixed conf. 0.25	0.25	24.56	32.46	12.94	+23.28	+11.06
960	Validation calibrated	0.50	15.72	22.26	8.79	-7.42	-3.53
960	Test diagnostic best	0.40	13.31	19.64	8.17	+4.83	+2.30
1280	Fixed conf. 0.25	0.25	21.23	30.07	11.41	+20.23	+9.61
1280	Validation calibrated	0.50	14.01	20.49	7.98	-4.55	-2.16
1280	Test diagnostic best	0.40	12.70	19.62	7.59	+5.38	+2.56

At fixed confidence 0.25, 1280 pixels gave the lowest MAE but all resolutions showed substantial positive bias, indicating over-counting. Under the formal validation-calibrated protocol, 800 pixels achieved the lowest MAE and near-zero total-count bias. In the diagnostic test-set sweep, 800 pixels remained lowest in MAE, while 1280 pixels was close. These results show that counting conclusions depend on the threshold protocol and should be interpreted separately from ranking-based AP.

In Eqs. (1)-(3), $A(r)$ denotes test mAP50-95 at resolution r , and $C_c(r)$ denotes the cost value for criterion c , with minima and maxima computed over the candidate set R . Accuracy is normalized as a benefit metric, whereas training time, GFLOPs, total latency, and peak reserved memory are converted into efficiency scores so that lower cost gives a larger normalized value. If a criterion has zero range across candidate resolutions, all candidates are assigned the same normalized score for that criterion. Equation (3) is not a universal utility function; it is an explicit example of task-objective-dependent resolution selection.

Table 10 Deployment and multi-objective resolution-selection summary

Res. (px)	Test mAP50-95	Train (h)	GFLOPs	Total latency (ms/img)	Peak reserved (MiB)	Score	Rank
800	0.7199	4.48	10.06	6.32	86	0.500	2
960	0.7420	7.29	14.49	8.52	102	0.656	1
1280	0.7543	22.54	25.76	9.15	142	0.500	2

Note: Latency and memory values were obtained from the 30-repeat locked-weight validation benchmark in Table S14. The 800- and 1280-pixel settings had equal default multi-objective scores and were therefore assigned the same tie rank. The score is objective-dependent and is not intended as a universal utility function.

The 30-repeat locked-weight recheck showed a monotonic increase in median total latency from 6.32 ms/image at 800 pixels to 8.52 ms/image at 960 pixels and 9.15 ms/image at 1280 pixels. The 1280-pixel setting retained the highest accuracy but also required the highest GFLOPs and peak reserved CUDA memory. Under the explicit multi-objective weighting scheme and the three tested resolutions, the 960-pixel setting obtained the highest default score (0.656), whereas the 800- and 1280-pixel settings obtained equal default scores (0.500) for different reasons: 800 pixels had the lowest cost but lowest accuracy, while 1280 pixels had the highest accuracy but highest cost. Thus, 960 pixels is not inherently best; it is the balanced candidate selected by the stated weighting scheme.

$$\text{Score}_\alpha(r) = \alpha N_{\text{acc}}(r) + (1 - \alpha)[0.4 N_{\text{train}}(r) + 0.2 N_{\text{GFLOPs}}(r) + 0.2 N_{\text{latency}}(r) + 0.2 N_{\text{memory}}(r)]. \quad (4)$$

The weight-sensitivity analysis in Table S16 used Eq. (4), where α controls the emphasis on accuracy and the internal cost-efficiency mixture keeps the relative ratio Training:GFLOPs:Latency:Memory = 0.4:0.2:0.2:0.2. When α was below 0.340, the cost-prioritized objective selected 800 pixels; when $0.340 \leq \alpha \leq 0.652$, the balanced objective selected 960 pixels; and when α exceeded 0.652, the accuracy-prioritized objective selected 1280 pixels. The 960-pixel result should therefore be interpreted as a protocol-derived balanced candidate under the tested resolution set and a moderate accuracy-prioritized objective range, not as a universal knee point or inherently optimal input resolution.

Seed repeatability across three input resolutions

YOLO11n seed-repeatability results across all three input resolutions are reported in Table 11. The main ablation table used seed 42, and additional repetitions were conducted using seeds 123 and 2024. The same leakage-controlled split and evaluation protocol were maintained across all runs.

Table 11 YOLO11n seed-repeatability check across three input resolutions

Resolution	Seed	Val mAP50	Val mAP50-95	Test mAP50	Test mAP50-95	Training time (h)
800	42	0.975	0.739	0.9643	0.7199	4.480
800	123	0.975	0.741	0.9656	0.7244	5.364
800	2024	0.976	0.746	0.9650	0.7220	5.573
960	42	0.984	0.768	0.9739	0.7420	7.290
960	123	0.984	0.770	0.9730	0.7360	7.555*
960	2024	0.984	0.769	0.9730	0.7400	7.200
1280	42	0.985	0.782	0.9741	0.7543	22.540
1280	123	0.986	0.783	0.9744	0.7559	24.189
1280	2024	0.984	0.784	0.9730	0.7530	14.863†

Note: The main ablation table uses seed=42. * The 960-pixel seed=123 training time was estimated from run-directory timestamps because the original console completion line was unavailable. † The 1280-pixel seed=2024 run early-stopped at epoch 179 of 300, with the best model observed at epoch 129 under patience=50; this indicates no further improvement in the validation fitness used for checkpoint selection within the patience window.

The trend remained stable across repeated runs. Test mAP50-95 ranged from 0.7199 to 0.7244 at 800 pixels, from 0.7360 to 0.7420 at 960 pixels, and from 0.7530 to 0.7559 at 1280 pixels. The relative ordering of the three resolutions was preserved across seeds, with 1280 pixels remaining the accuracy-oriented setting and 960 pixels consistently improving over 800 pixels. The 1280-pixel seed-2024 run triggered early stopping at epoch 179 of 300, with the best model selected from epoch 129; this run was retained because its test result was consistent with the other 1280-pixel repetitions.

Cross-detector robustness check with YOLOv8n

To examine whether the observed resolution trend was specific to YOLO11n, YOLOv8n was additionally evaluated as a cross-detector robustness check under the same leakage-controlled split and input-resolution protocol. This experiment was not intended as a detector-ranking benchmark, but as an auxiliary check of whether the resolution-diagnosis trend persisted across another lightweight YOLO-family detector.

YOLOv8n showed the same monotonic resolution-dependent trend as YOLO11n. Test mAP50-95 increased from 0.7272 at 800 pixels to 0.7443 at 960 pixels and 0.7573 at 1280 pixels. This supports that the proposed resolution-selection protocol was not tied to a single YOLO11n training run or detector variant. The complete YOLOv8n cross-detector results are reported in Table S13.

External directional protocol-transfer demonstration on VisDrone2019-DET

To examine whether the proposed input-resolution selection protocol could be applied outside the broiler-monitoring domain, an external directional protocol-transfer demonstration was conducted on VisDrone2019-DET (Zhu et al. 2022). This experiment was not a VisDrone state-of-the-art benchmark or a fully matched second primary benchmark. Instead, it tested whether the same resolution-diagnosis workflow could be applied to a multi-class aerial dense small-object dataset under a practical compute budget.

The VisDrone experiment used 100 epochs because it served as an external directional demonstration rather than a full repetition of the primary PIO protocol. The VisDrone training split contains 6,471 images, substantially more than the PIO training split; therefore, the reduced epoch count was intended to test directional protocol transfer under a practical compute budget. The official VisDrone splits were

audited for exact duplicates and SSIM-confirmed near duplicates, but they were not reconstructed using the source-group splitting procedure used for the primary PIO case study.

YOLO11n was trained for 100 epochs at 800, 960, and 1280 pixels using seed 42, batch size 2, and SGD. Absolute AP values were substantially lower than those obtained on PIO, as expected from the more difficult multi-class aerial setting, smaller objects, heavier occlusion, larger viewpoint variation, and more diverse backgrounds. Nevertheless, the same directional resolution trend was observed on both validation and test-dev splits. On the test-dev split, mAP50-95 increased from 0.1783 at 800 pixels to 0.2007 at 960 pixels and 0.2300 at 1280 pixels. The validation split showed the same ordering, increasing from 0.2180 to 0.2488 and 0.2885. Complete results are reported in Table S17.

An auxiliary duplicate audit of the official VisDrone train/validation/test-dev splits found zero cross-split basename overlaps, zero cross-split image-MD5 duplicate pairs, zero cross-split annotation-MD5 duplicate pairs, and zero SSIM-confirmed near-duplicate pairs. Six dHash candidates were identified, but none reached the SSIM ≥ 0.95 confirmation threshold. These results support the protocol-transfer interpretation but do not prove a universal resolution law across dense small-object datasets.

Discussion

The results support a protocol-oriented interpretation rather than a detector-oriented claim. The goal was not to introduce a new detector, loss function, or network component, but to provide a reproducible procedure for evaluating, diagnosing, and selecting input resolution under leakage control and task-specific deployment objectives. The central question is not which of 800, 960, and 1280 pixels is universally best, but how input resolution should be selected once the target objective, leakage risk, subgroup behavior, counting behavior, uncertainty, and deployment cost are made explicit.

For precision poultry monitoring, the practical value of the protocol is that it connects computer-vision metrics with farm-deployment decisions. The same detection model may support flock counting, density monitoring, distribution assessment, welfare-related observation, and near-real-time camera-based surveillance. These objectives do not always favor the same input resolution. A system that prioritizes scalar counting or low-cost edge deployment may prefer a lower resolution, whereas a system that requires more accurate localization of individual birds may justify higher computational cost.

The protocol is therefore a decision sequence rather than a fixed recommendation. A user first audits the split for source-level and near-duplicate leakage, then evaluates candidate resolutions under identical detector and training conditions, and finally interprets the results through the target objective. In the present case, the same experiments support three different decisions: 800 pixels for validation-calibrated scalar counting, 1280 pixels for maximum box-level AP, and 960 pixels only under the specified accuracy-cost weighting.

Strict global-first diagnostic AP should be treated as an auxiliary diagnostic measure, not as a replacement for ordinary AP or standard global AP. Its purpose is to preserve the global prediction ranking and matching process before attributing matched detections to scale- or density-defined subgroups. It is useful for examining how subgroups contribute under the full-test-set operating order, but the main performance conclusions are based on ordinary AP, standard global AP, seed-repeatability results, the YOLOv8n cross-detector check, and bootstrap uncertainty analysis.

The results do not support treating 960 pixels as a universally preferable deployment resolution. If the objective is scalar image-level counting after validation-based calibration, 800 pixels may be preferable in

this dataset. If the objective is box-level detection and localization, 1280 pixels achieved the highest global and ordinary subgroup AP. If the objective jointly considers AP, training cost, GFLOPs, latency, and memory pressure under the specified weighting scheme, 960 pixels is selected as a balanced candidate. The weight-sensitivity analysis further showed that 800 pixels was selected under cost-prioritized objectives, 960 pixels when $0.340 \leq \alpha \leq 0.652$, and 1280 pixels under strongly accuracy-prioritized objectives.

The 960-pixel interpretation is limited to the three tested discrete resolutions and the default multi-objective weighting scheme. It should not be interpreted as evidence that 960 pixels is superior to untested intermediate resolutions such as 1024 or 1120 pixels.

The distinction between detection AP and image-level counting error is important for deployment. AP is integrated over confidence thresholds and IoU levels, whereas counting error is evaluated at a fixed threshold after validation-based calibration. A counting-only objective may therefore prefer a different resolution from a localization-oriented detection objective, reinforcing the need to report MAE, RMSE, MAPE, and bias separately from AP.

The deployment benchmark also shows why resolution selection should not be summarized by inference time alone. In the 30-repeat locked-weight FP16 validation benchmark, total latency increased monotonically from 800 to 960 and 1280 pixels. However, the largest engineering penalties of 1280 pixels were its training time, GFLOPs, and peak reserved CUDA memory. Deployment-oriented resolution selection should therefore jointly consider training cost, computational load, latency, memory pressure, and task-specific accuracy requirements.

The consistent resolution trend on PIO and VisDrone should not be interpreted as a new general law of small-object detection. Higher input resolution is expected to benefit small-object localization in many settings. The contribution here is the reproducible protocol used to quantify accuracy, subgroup behavior, counting behavior, uncertainty, and deployment cost under explicit audit and reporting rules. The VisDrone experiment supports directional protocol transfer to a non-agricultural dense small-object benchmark, but it does not prove universal cross-dataset robustness.

This study has several limitations. First, the validation subset contained only two source components, with the largest validation component contributing 130 of 153 images. This constraint was examined through the added source-component sensitivity analysis (Supplementary Table S18). Full-validation and dominant-component calibration selected identical thresholds for all three resolutions, whereas the smaller component changed selected thresholds, showing that counting calibration is partly sensitive to validation-source composition. Second, the PIO dataset was the primary agricultural dataset, so the numerical resolution preference should be tested on additional poultry houses, viewpoints, stocking densities, and livestock species. Third, only three discrete resolutions were evaluated, so 960 pixels should not be interpreted as an inherent optimum among untested intermediate input sizes. Fourth, the external VisDrone experiment used one detector, one seed, and the official benchmark splits and should be interpreted only as a directional protocol-transfer demonstration. Fifth, the batch size of 2 reflected the resource-constrained 8 GB GPU setting and was fixed across resolutions to preserve a controlled comparison. Sixth, the deployment benchmark was conducted under a specific hardware and software stack, and absolute latency may differ under TensorRT, ONNX Runtime, embedded GPUs, or optimized NMS implementations. Finally, the early-stopped 1280-pixel seed-2024 run suggests that high-resolution convergence time may depend on seed-specific validation dynamics.

Conclusions

This study introduced a leakage-controlled and deployment-aware input-resolution selection protocol for dense broiler detection in precision poultry monitoring. It did not propose a new detector architecture. The first contribution is an auditable evaluation workflow based on exact-duplicate removal, source-group binding, density-aware grouped allocation, source-component reporting, and cross-split leakage audits. This workflow reduces the risk that apparent resolution gains are inflated by near-duplicate samples, source-related leakage, or density imbalance.

The second contribution is a detection-counting-deployment evaluation framework for high-density poultry scenes. Resolution effects were analyzed using standard global AP, scale-stratified AP, density-stratified AP, strict global-first diagnostic AP, TP/FN/FP summaries, counting errors, bootstrap uncertainty, and deployment indicators including training time, GFLOPs, latency, and validation/inference memory pressure. Strict diagnostic AP is used only as an auxiliary diagnostic tool; the main performance conclusions are based on ordinary AP and global AP.

The third contribution is a task-objective-driven resolution-selection rule with explicit normalization, weighted scoring, and weight-sensitivity analysis. The results should not be summarized as a claim that 960 pixels is generally best. Under the tested resolutions, 1280 pixels was the accuracy-oriented setting, 800 pixels was the lowest-cost baseline and the best validation-calibrated scalar-counting setting in this test set, and 960 pixels was selected only as a balanced candidate under the specified multi-objective weighting scheme. Overall, the study provides a reproducible method for choosing input resolution in dense broiler-monitoring systems under leakage control, counting calibration, uncertainty assessment, and explicit deployment objectives.

Supplementary information

The following supplementary materials are provided as Online Resources.

Online Resource 1 Cross-split leakage audit and split diagnostics. This resource includes the cross-split leakage audit, near-duplicate audit using pHash/dHash and SSIM, component-level split diagnostics, and dataset split statistics.

Online Resource 2 Maximum-detection sensitivity and strict diagnostic AP analysis. This resource includes the full max_det sensitivity results, strict global-first scale diagnostic AP, strict global-first density diagnostic AP, and TP/FN/FP summaries by density group.

Online Resource 3 Counting calibration and confidence-threshold sensitivity. This resource includes fixed-threshold counting at confidence threshold 0.25, density-specific fixed-threshold counting, test-set confidence sweep for diagnostic purposes, and validation confidence-MAE sweep curves.

Online Resource 4 Seed repeatability, cross-detector robustness, deployment benchmarking, multi-objective resolution selection, external directional protocol-transfer demonstration, and qualitative examples. This resource includes the three-seed YOLO11n repeatability check across all three resolutions, the YOLOv8n cross-detector robustness check, the 30-repeat locked-weight deployment benchmark, the multi-objective resolution-selection analysis, the weight-sensitivity analysis, the VisDrone2019-DET directional cross-domain demonstration, and representative low- and medium-density qualitative cases where increased input resolution did not provide a clear visual advantage.

Online Resource 5 Bootstrap delta-difference analysis. This resource includes paired image-level bootstrap AP samples, bootstrap delta summaries, and delta-difference analysis for comparing the marginal gain from 800 to 960 and from 960 to 1280.

Statements and declarations

Funding

No funding was received for conducting this study.

Competing interests

The authors declare no competing interests that are relevant to the content of this article.

Ethics approval

Not applicable. This study used existing image datasets and did not involve new animal experiments conducted by the authors.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Author contributions

Y.S. contributed to study conception and design, data curation, split construction, leakage auditing, model training, evaluation, visualization, analysis, and first-draft preparation. J.X. and Z.D. supervised the study and contributed to conceptual guidance, manuscript revision, and correspondence. H.L. contributed to manuscript review, discussion, and refinement. All authors read and approved the final manuscript.

Data availability

The original PIO dataset is publicly available from its official source (Boniche et al. 2026). The VisDrone2019-DET dataset used for the external directional protocol-transfer demonstration is publicly available from the official VisDrone source (Zhu et al. 2022). The leakage-controlled split manifest, source-group metadata, leakage-audit summaries, group definitions, bootstrap outputs, counting-calibration outputs, supplementary source data, evaluation scripts, traceability logs, fixed figure source data, VisDrone conversion script, and VisDrone auxiliary duplicate-audit outputs generated for this study are available in the public GitHub repository: <https://github.com/uaena7-bit/pio-resolution-diagnosis-yolo11n>. The archived version corresponding to this manuscript is GitHub release tag v1.1.0 (Song 2026).

These materials support reproduction and inspection of the dataset split, leakage checks, resolution-selection protocol, scale- and density-stratified analyses, bootstrap analysis, counting-calibration results, Tables S4-S17 source data, source data for Figs. 2-4 and Figs. S2-S3, Fig. 5 qualitative case metadata,

Fig. S1 qualitative-case metadata, and the final figure-generation scripts. Raw PIO and VisDrone images are not redistributed by the authors and should be obtained from the original dataset sources. Qualitative figure scripts require local access to the original images and annotations.

Code availability

The split-construction, evaluation, deployment-benchmarking, multi-objective scoring, VisDrone conversion and audit, and figure-generation scripts are available in the same GitHub repository, release v1.1.0 (Song 2026). The release includes scripts for leakage-controlled split construction, plotting from fixed source data, qualitative-figure metadata and visualization, strict diagnostic AP plotting, validation confidence-MAE sweeps, YOLOv8n cross-detector source-data preparation, 30-repeat locked-weight YOLO11n deployment benchmarking, multi-objective scoring, weight-sensitivity analysis, VisDrone2019-DET conversion, and VisDrone split-audit reporting.

Reproducibility checklist

The following checklist summarizes the main reproducibility materials provided with the manuscript and public release.

Item	Availability	Notes
Fixed split manifest	Provided	PIO-GRDB-MD5-7_1_2 train/validation/test split
Leakage audit summaries	Provided	Source path, filename, MD5, and source-group checks
Near-duplicate audit configuration	Provided	pHash/dHash candidate search and SSIM verification
Source-group definitions	Provided	Used for grouped split construction
YOLO11n evaluation scripts	Provided	Main resolution diagnosis and seed-repeatability evaluation
YOLOv8n robustness-check data	Provided	Table S13 source data
Deployment benchmark data	Provided	Table S14 30-repeat source data and raw repeats
Multi-objective score data	Provided	Tables S15 and S16 source data and README
Figure source data	Provided	Main and supplementary figure source CSV files
Raw PIO images	Not redistributed	Require authorized dataset access from the original source
VisDrone cross-domain data	Provided	Table S17 source data after obtaining VisDrone
VisDrone auxiliary audit	Provided	Official split duplicate audit and near-duplicate candidates
Raw VisDrone images	Not redistributed	Require official VisDrone access

References

Akyon FC, Altinuc SO, Temizel A (2022) Slicing aided hyper inference and fine-tuning for small object detection. arXiv:2202.06934. Preprint.

Bhujel A, Wang Y, Lu Y, Morris D, Dangol M (2025) A systematic survey of public computer vision datasets for precision livestock farming. Computers and Electronics in Agriculture 229:109718. <https://doi.org/10.1016/j.compag.2024.109718>

Biswas R, Blanco-Medina P (2021) State of the art: Image hashing. arXiv:2108.11794. Preprint.

Bochkovskiy A, Wang C-Y, Liao H-YM (2020) YOLOv4: Optimal speed and accuracy of object detection. arXiv:2004.10934. Preprint.

Bodla N, Singh B, Chellappa R, Davis LS (2017) Soft-NMS: Improving object detection with one line of code. In: Proceedings of the IEEE International Conference on Computer Vision, pp 5561-5569. <https://doi.org/10.1109/ICCV.2017.593>

Boniche K, Cruz E, Rangel JC, Hidalgo-Rodríguez M, Gómez-Donoso F (2026) PIO, a large-scale dataset for broiler chicken detection under real poultry farming conditions. *Scientific Data* 13:801. <https://doi.org/10.1038/s41597-026-07114-5>

Cheng G, Yuan X, Yao X, Yan K, Zeng Q, Xie X, Han J (2022) Towards large-scale small object detection: Survey and benchmarks. arXiv:2207.14096. Preprint.

Efron B, Tibshirani RJ (1993) *An Introduction to the Bootstrap*. Chapman & Hall/CRC, New York. <https://doi.org/10.1007/978-1-4899-4541-9>

Feng Q, Xu X, Wang Z (2023) Deep learning-based small object detection: A survey. *Mathematical Biosciences and Engineering* 20(4):6551-6590. <https://doi.org/10.3934/mbe.2023282>

Guo Y, Aggrey SE, Wang P, Oladeinde A, Chai L (2022) Monitoring behaviors of broiler chickens at different ages with deep learning. *Animals* 12(23):3390. <https://doi.org/10.3390/ani12233390>

Guo Y, Aggrey SE, Yang X, Oladeinde A, Qiao Y, Chai L (2023) Detecting broiler chickens on litter floor with the YOLOv5-CBAM deep learning model. *Artif. Intell. Agric.* 9:36-45. <https://doi.org/10.1016/j.aiia.2023.08.002>

Guo Y, Wu Z, You B, Chen L, Zhao J, Li X (2025) YOLO-SDD: An effective single-class detection method for dense livestock production. *Animals* 15(9):1205. <https://doi.org/10.3390/ani15091205>

Jocher G, Qiu J (2024) Ultralytics YOLO11. Ultralytics software. <https://github.com/ultralytics/ultralytics>. Accessed 22 May 2026.

Joeres R, Blumenthal DB, Kalinina OV (2025) Data splitting to avoid information leakage with DataSAIL. *Nature Communications* 16:3337. <https://doi.org/10.1038/s41467-025-58606-8>

Li G (2025) A survey of open-access datasets for computer vision in precision poultry farming. *Poultry Science* 104(2):104784. <https://doi.org/10.1016/j.psj.2025.104784>

Li K, Wan G, Cheng G, Meng L, Han J (2020) Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing* 159:296-307. <https://doi.org/10.1016/j.isprsjprs.2019.11.023>

Lin T-Y, Dollar P, Girshick R, He K, Hariharan B, Squeeze B (2017) Feature pyramid networks for object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 936-944. <https://doi.org/10.1109/CVPR.2017.106>

Lin T-Y, Maire M, Belongie S, Hays J, Perona P, Ramanan D, Dollar P, Zitnick CL (2014) Microsoft COCO: Common objects in context. In: *Computer Vision - ECCV 2014*, pp 740-755. Springer, Cham. https://doi.org/10.1007/978-3-319-10602-1_48

Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu C-Y, Berg AC (2016) SSD: Single Shot MultiBox Detector. In: Computer Vision - ECCV 2016, pp 21-37. Springer, Cham. https://doi.org/10.1007/978-3-319-46448-0_2

Lv J, Wang J, Peng C, Huang Q (2023) DFCCNet: A dense flock of chickens counting network based on density map regression. *Animals* 13(23):3729. <https://doi.org/10.3390/ani13233729>

Neethirajan S (2022) ChickTrack: A quantitative tracking tool for measuring chicken activity. *Measurement* 191:110819. <https://doi.org/10.1016/j.measurement.2022.110819>

Nikouei M, Baroutian B, Nabavi S, Taraghi F, Aghaei A, Sajedi A, Moghaddam ME (2025) Small object detection: A comprehensive survey on challenges, techniques and real-world applications. *Intelligent Systems with Applications* 27:200561. <https://doi.org/10.1016/j.iswa.2025.200561>

Padilla R, Passos WL, Dias TLB, Netto SL, da Silva EAB (2021) A comparative analysis of object detection metrics with a companion open-source toolkit. *Electronics* 10(3):279. <https://doi.org/10.3390/electronics10030279>

Redmon J, Divvala S, Girshick R, Farhadi A (2016) You Only Look Once: Unified, real-time object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 779-788. <https://doi.org/10.1109/CVPR.2016.91>

Redmon J, Farhadi A (2017) YOLO9000: Better, faster, stronger. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 6517-6525. <https://doi.org/10.1109/CVPR.2017.690>

Redmon J, Farhadi A (2018) YOLOv3: An incremental improvement. arXiv:1804.02767. Preprint.

Ren S, He K, Girshick R, Sun J (2017) Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39(6):1137-1149. <https://doi.org/10.1109/TPAMI.2016.2577031>

Rohan A, Rafaq MS, Hasan MJ, Asghar F, Bashir AK, Dottorini T (2023) Application of deep learning for livestock behaviour recognition: A systematic literature review. arXiv:2310.13483. Preprint.

Song Y (2026) Reproducibility materials for leakage-controlled input-resolution diagnosis of dense small-object detection on PIO. GitHub repository, release v1.1.0. <https://github.com/uaena7-bit/pio-resolution-diagnosis-yolo11n/releases/tag/v1.1.0>. Accessed 25 Jun 2026.

Tan M, Pang R, Le QV (2020) EfficientDet: Scalable and efficient object detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 10781-10790. <https://doi.org/10.1109/CVPR42600.2020.01079>

van der Eijk JAJ, Guzhva O, Voss A, Moeller M, Giersberg MF, Jacobs L, de Jong IC (2022) Seeing is caring - automated assessment of resource use of broilers with computer vision techniques. *Frontiers in Animal Science* 3:945534. <https://doi.org/10.3389/fanim.2022.945534>

Wang A, Chen H, Liu L, Chen K, Lin Z, Han J, Ding G (2024) YOLOv10: Real-time end-to-end object detection. arXiv:2405.14458. Preprint.

Wang C-Y, Bochkovskiy A, Liao H-YM (2021) Scaled-YOLOv4: Scaling cross stage partial network. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 13024-13033. <https://doi.org/10.1109/CVPR46437.2021.01283>

Wang C-Y, Bochkovskiy A, Liao H-YM (2023) YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 7464-7475. <https://doi.org/10.1109/CVPR52729.2023.00721>

Wang C-Y, Yeh I-H, Liao H-YM (2024) YOLOv9: Learning what you want to learn using programmable gradient information. arXiv:2402.13616. Preprint.

Wang Z, Bovik AC, Sheikh HR, Simoncelli EP (2004) Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing 13(4):600-612.
<https://doi.org/10.1109/TIP.2003.819861>

Zhu P, Wen L, Bian X, Ling H, Hu Q (2018) Vision meets drones: A challenge. arXiv:1804.07437. Preprint.

Zhu P, Wen L, Du D, Bian X, Fan H, Hu Q, Ling H (2022) Detection and tracking meet drones challenge. IEEE Transactions on Pattern Analysis and Machine Intelligence 44(11):7380-7399.
<https://doi.org/10.1109/TPAMI.2021.3119563>